

JOKIOINEN, Finland

WMO: 029630

Lat: 60.814N

Lon: 23.497E

Elev: 103

StdP: 100.09

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.2	-19.7	-25.5	0.4	-23.1	-22.0	0.5	-19.4	9.7	-2.0	8.8	-2.1	1.9	60	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	26.7	18.1	24.9	17.2	23.1	16.4	19.7	24.3	18.5	23.0	17.4	21.6	3.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.1	21.6	16.8	12.1	20.3	15.7	11.3	19.1	56.6	24.4	52.8	23.2	49.4	21.7	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 8.7	7.8	7.0														
(5)			DB	-26.1	28.8	3.6	2.2	-28.7	30.4	-30.8	31.7	-32.8	32.9	-35.4	34.6	
			WB	-26.3	20.7	3.5	1.4	-28.8	21.8	-30.9	22.6	-32.9	23.4	-35.5	24.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-5.5	-5.9	-2.3	3.9	9.5	13.9	16.9	15.4	10.8	5.0	0.6	-2.9
	DBStd	9.08	6.31	6.53	4.64	3.76	4.05	3.26	2.98	3.01	3.37	3.94	4.50	5.77
	HDD10.0	2428	482	444	381	185	58	4	0	1	31	159	283	400
	HDD18.3	4895	740	678	639	432	274	137	66	98	227	413	533	658
	CDD10.0	609	0	0	0	4	43	122	213	170	54	4	0	0
	CDD18.3	34	0	0	0	0	1	5	20	8	0	0	0	0
	CDH23.3	351	0	0	0	0	19	60	201	71	0	0	0	0
	CDH26.7	51	0	0	0	0	2	6	37	7	0	0	0	0
(14) Wind	WSAvg	3.5	3.6	3.6	3.7	3.5	3.5	3.4	3.1	3.2	3.3	3.6	3.6	3.8
(15) Precipitation	PrecAvg	593	41	28	27	31	36	58	77	77	59	60	55	46
	PrecMax	787	81	73	52	83	87	131	164	184	136	154	128	92
	PrecMin	373	6	1	1	2	1	11	1	13	12	13	3	0
	PrecStd	86	19	16	13	17	19	31	39	35	30	31	25	22
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	5.1	10.5	19.3	25.7	27.4	29.8	27.6	21.5	14.3	9.6	7.6
		MCWB	4.2	3.3	5.6	10.9	15.4	17.7	19.5	18.6	15.6	11.6	9.0	7.0
	2%	DB	3.4	3.2	7.1	16.2	22.8	25.1	27.6	25.2	19.5	12.7	8.5	6.2
		MCWB	2.8	2.4	3.6	9.2	14.1	16.6	19.1	17.4	14.8	11.2	7.9	5.7
	5%	DB	2.4	2.2	4.9	13.5	20.3	23.0	25.7	23.5	17.7	11.7	7.4	4.8
		MCWB	2.0	1.6	2.5	7.8	12.8	15.7	18.2	16.9	13.6	10.6	6.9	4.3
	10%	DB	1.4	1.4	3.1	11.0	17.7	20.9	23.6	21.6	16.1	10.7	6.4	3.6
		MCWB	1.0	0.9	1.5	6.5	11.2	14.4	17.3	16.1	13.3	9.7	5.9	3.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	3.6	6.1	11.9	17.5	19.7	21.7	20.2	16.8	12.7	9.2	7.1
		MCDB	4.6	4.4	9.6	17.3	23.2	24.5	26.6	24.8	19.4	13.4	9.5	7.4
	2%	WB	2.9	2.6	4.0	9.9	15.3	18.0	20.3	18.9	15.6	11.7	8.0	5.8
		MCDB	3.2	3.1	6.3	14.9	20.5	23.3	25.1	22.9	18.1	12.4	8.3	6.1
	5%	WB	2.0	1.8	2.8	8.4	13.6	16.5	19.1	17.8	14.6	10.8	7.0	4.4
		MCDB	2.4	2.2	4.3	12.6	18.6	21.1	23.8	21.6	16.7	11.5	7.4	4.7
	10%	WB	1.1	1.0	1.8	6.9	11.9	15.2	18.0	16.8	13.7	9.8	5.9	3.1
		MCDB	1.4	1.4	2.9	10.6	17.0	19.7	22.5	20.3	15.6	10.5	6.3	3.5
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.8	7.5	8.9	10.5	9.4	9.6	9.6	8.0	5.6	3.9	4.5
		MCDBR	3.7	4.9	8.8	13.5	14.7	13.1	13.2	12.3	10.1	5.9	4.0	3.9
	5% WB	MCWBR	3.7	4.4	6.0	7.7	7.2	6.2	5.8	5.6	5.8	4.4	3.9	3.7
		MCDWR	3.9	4.4	7.5	12.1	12.2	10.9	11.1	9.8	8.3	5.0	4.0	3.7
(40) Clear-Sky Solar Irradiance	taub		0.261	0.300	0.326	0.363	0.359	0.354	0.379	0.369	0.341	0.325	0.288	0.377
		taud	2.108	2.182	2.265	2.344	2.416	2.451	2.398	2.422	2.493	2.407	2.202	2.634
	Edn at Noon		469	673	792	831	866	876	839	819	782	656	435	322
		Edh at Noon	41	70	91	102	102	101	104	94	74	57	38	27
(44) All-Sky Solar Radiation	RadAvg		0.23	0.81	2.29	3.80	5.17	5.44	5.12	4.02	2.45	1.06	0.29	0.14
	RadStd		0.04	0.17	0.28	0.40	0.52	0.44	0.52	0.43	0.31	0.19	0.06	0.01

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		+0.48	N/A	N/A	N/A	N/A	N/A	-147	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A	N/A	N/A

Nomenclature: See separate page